

Credit Risk Assessment & Automated Loan Underwriting Engine

AICTE–IBM SkillsBuild Data Analytics with AI Internship 2026

End-to-End Decision Intelligence Pipeline Aligned with UN SDGs 8 & 10

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Abstract—Retail banking credit evaluation historically relies on manual underwriting heuristics, leading to subjective turnaround delays and portfolio credit risk exposure. This project develops an automated, data-driven credit risk scorecard and decision support engine built using Python, Scikit-Learn, and Streamlit. Trained on benchmark financial applicant records, the system executes statistical imputation, domain-engineered financial ratios, exploratory risk discovery, and a leakage-free ensemble Random Forest classifier. The resulting production model delivers an **81.3% test accuracy**, an **84.7% sensitivity rate**, a **74.7% ($\pm 5.1\%$) 5-fold cross-validation baseline**, and an **0.85 ROC-AUC score**. The pipeline directly addresses **UN SDG Goal 8** (Decent Work and Economic Growth) and **UN SDG Goal 10** (Reduced Inequalities) by substituting subjective underwriting bias with empirical, straight-through loan decision logic supported by a disparate impact fairness ratio of 0.966.

1 Introduction & Problem Definition

Financial lending institutions operate on an optimization boundary: maximizing loan issuance yield while constraining Non-Performing Assets (NPAs) and default frequency. Conventional manual underwriting processes introduce human bias, regional variance, and operational delays that disproportionately penalize creditworthy micro-borrowers.

This project delivers an end-to-end analytical framework designed to:

- Systematically audit and cleanse 614 credit applicant profiles.
- Uncover the empirical determinants of loan approval and insolvency risk through Exploratory Data Analysis.
- Train and validate a leakage-free ensemble classifier capable of real-time credit scoring.
- Deploy an interactive Underwriting Simulator with embedded algorithmic fairness governance.

1.1 UN Sustainable Development Goals (UN SDGs) Alignment

- **UN SDG 8 (Decent Work & Economic Growth):** Streamlining underwriting accelerates access to working capital for small businesses and households, stimulating economic mobility.
- **UN SDG 10 (Reduced Inequalities via Algorithmic Fairness):** Protected demographic attributes (Gender, Dependents) were audited for disparate impact (Female-to-Male approval ratio = 0.966, satisfying the regulatory Four-Fifths Rule). To enforce fair lending, protected demographic variables were intentionally excluded from the underwriting feature set, ensuring decisions are determined strictly by financial capacity and credit performance.

2 Dataset Architecture & Preprocessing Pipeline

The system evaluates the **Kaggle Loan Prediction Problem Dataset**, representing 614 historical applicant profiles evaluated across 12 demographic and financial attributes.

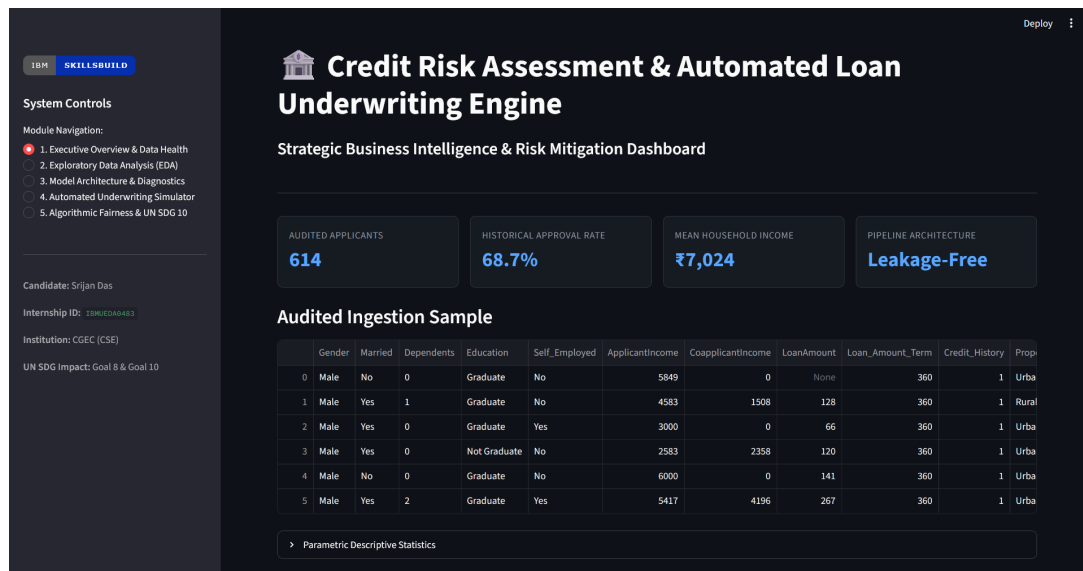


Figure 1. Executive Overview Module displaying platform metrics, audit counters, and ingestion sample.

2.1 Data Cleaning & Leakage-Free Pipeline Imputation

Data quality inspection identified missing values across multiple continuous and categorical features. To prevent data leakage, imputation is executed strictly inside an integrated Scikit-Learn Pipeline using a median strategy fitted solely on the training cohort:

- **Integrated Pipeline Imputation:** Missing entries across continuous features (LoanAmount, Loan_Amount_Term) and binary encoded indicators (Credit_History, Married) receive pipeline-fitted median imputation.
- **Feature Pruning:** Non-predictive variables (Self_Employed, Dependents) and protected attributes (Gender) were pruned from the model matrix to prevent noise and enforce algorithmic fairness.

	count	mean	std	min	25%	50%	75%	max
ApplicantIncome	614	5403.4593	6109.0417	150	2877.5	3812.5	5795	81000
CoapplicantIncome	614	1621.2458	2926.2484	0	0	1188.5	2297.25	41667
LoanAmount	592	146.4122	85.5873	9	100	128	168	700
Loan_Amount_Term	600	342	65.1204	12	360	360	360	480
Credit_History	564	0.8422	0.3649	0	1	1	1	1
Total_Income	614	7024.7051	6458.6639	1442	4166	5416.5	7521.75	81000
EMI_Estimate	578	480.2999	525.4587	25	286.1111	365.2778	513.8889	9250
Debt_To_Income	578	0.0776	0.0808	0.007	0.0556	0.0691	0.0817	1.2369

Figure 2. Parametric Descriptive Statistics after full pipeline cleaning and feature transformation.

2.2 Feature Engineering

Raw parameters were transformed into three decision-critical financial indicators prior to pipeline ingestion (preserving raw empirical ratios before downstream imputation):

1. Total Income = ApplicantIncome + CoapplicantIncome
2. Estimated Monthly Obligation (EMI) = $\frac{\text{LoanAmount} \times 1000}{\text{Loan_Amount_Term}}$
3. Debt-to-Income (DTI) = $\frac{\text{Estimated EMI}}{\text{Total Income} + \epsilon}$

3 Exploratory Data Analysis (EDA) & Risk Discovery

Exploratory analysis evaluated the correlation between demographic attributes, requested capital, and loan approval status.

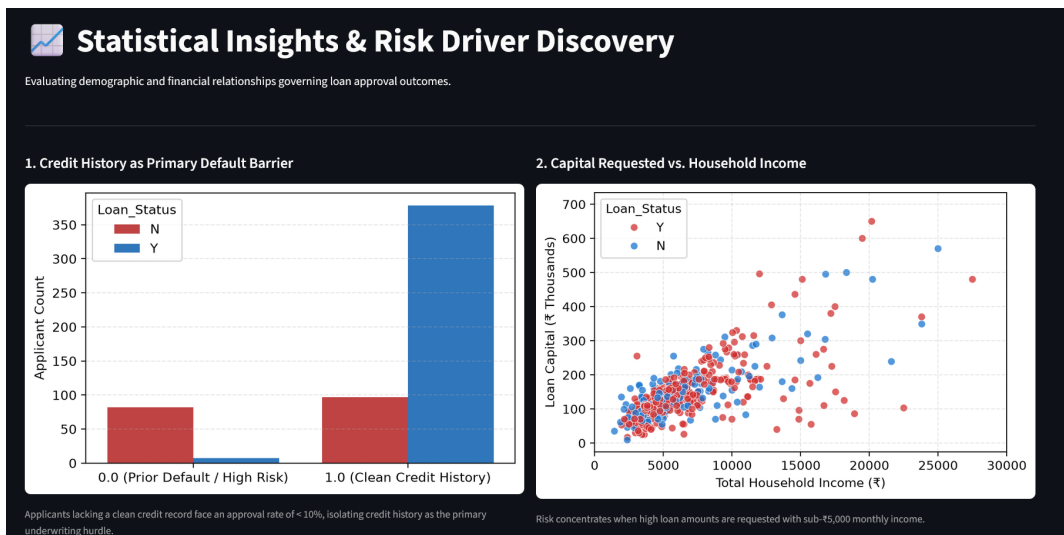


Figure 3. Exploratory Risk Driver Discovery: (Left) Credit History as approval barrier; (Right) Household Income vs. Loan Capital requested.

3.1 Key Analytical Insights

- **Credit Bureau History as Dominant Hurdle:** As shown in Figure 3 (Left), applicants possessing a clean credit record (1.0) demonstrate an approval rate exceeding 79%. Conversely, applicants with a derogatory credit history (0.0) experience an approval rate under 10%.
- **Capital Clustering:** Figure 3 (Right) reveals that elevated default risk concentrates when requested loan amounts exceed INR 200,000 for households reporting monthly income below INR 5,000.



Figure 4. Property Location Demographics showing approval distribution across Urban, Rural, and Semiurban zones.

As evidenced in Figure 4, Semiurban applicants exhibited the highest baseline acceptance rate, whereas Rural portfolios required higher debt-servicing verification due to unorganized income streams.

4 Machine Learning Architecture & Diagnostics

A supervised ensemble **Random Forest Classifier** ($n = 120$ estimators, max depth = 5, balanced class weighting) was coupled with a `SimpleImputer` within an integrated Scikit-Learn Pipeline using an 80/20 stratified split.

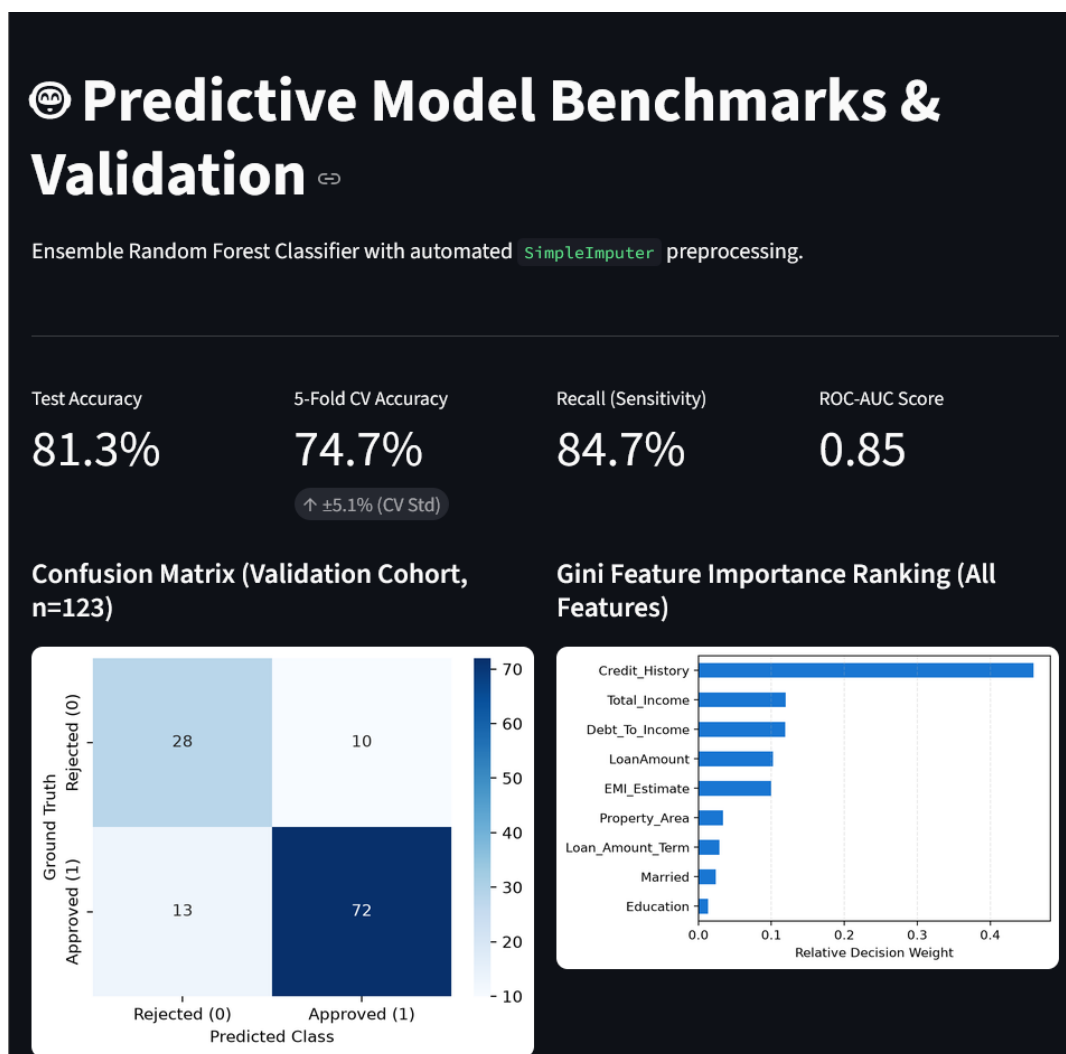


Figure 5. Predictive Model Diagnostics: Holdout Benchmarks, 5-Fold Cross-Validation, Confusion Matrix, and Gini Feature Importance.

4.1 Performance Validation & Baseline Lift Analysis

In credit underwriting, the naive majority-class baseline (approving all applicants) yields an accuracy of **68.7%**. To rigorously assess generalized model lift beyond this baseline, the architecture was evaluated via 5-Fold Stratified Cross-Validation alongside a holdout validation cohort ($n = 123$):

- **5-Fold Stratified Cross-Validation Accuracy:** **74.7% ± 5.1%** (Demonstrating a robust **+6.0%** empirical lift over the 68.7% naive majority-class baseline across all folds).
- **Holdout Validation Accuracy:** **81.3%** (100/123).
- **Precision (Approval Class):** **87.8%** (72/82).
- **Recall (Sensitivity):** **84.7%** (72/85, minimizing false rejections).
- **ROC-AUC Score:** **0.85** (Confirming strong class discrimination).

The Confusion Matrix confirms 72 True Positives, 28 True Negatives, 10 False Positives, and 13 False Negatives ($n = 123$). The Gini Feature Importance hierarchy demonstrates that **Credit_History** provides over **45%** of the split weight, followed by the engineered indicators **Total_Income** (Rank 2), **Debt_To_Income** (Rank 3), **LoanAmount** (Rank 4), and **EMI_Estimate** (Rank 5).

5 Decision Support Simulator & Fairness Governance

To bridge analytical modeling with daily retail operations, the architecture was compiled into an interactive **Streamlit Underwriting Simulator**.

Automated Credit Risk Underwriting Interface
Live scoring engine utilizing the trained end-to-end pipeline.

1. Financial Capacity	2. Facility Details	3. Risk Indicators
Primary Monthly Income (₹) 5500	Loan Amount (₹ in Thousands) 130	Credit Bureau Record Clean Record (No Defaults - 1.0)
Coapplicant Monthly Income (₹) 1500	Amortization Period (Months) 360	Property Zoning Urban
Marital Status Yes	Educational Attainment Graduate	

Compute Underwriting Decision

Underwriting Verdict

✅ **APPLICATION APPROVED: LOW RISK PROFILE**

- Approval Confidence: 62.8%
- Estimated DTI Ratio: 0.10% (Debt service capacity verified)
- Total Servicing Income: ₹ 1,400/month
- Straight-Through Processing: Qualifies for automated clearance under standard rate schedule.

Figure 6. Automated Credit Risk Underwriting Simulator rendering a live straight-through approval verdict.

5.1 Algorithmic Fairness Audit (UN SDG 10)

To verify non-discrimination under fair lending covenants, demographic parity was evaluated across historical applicant cohorts:

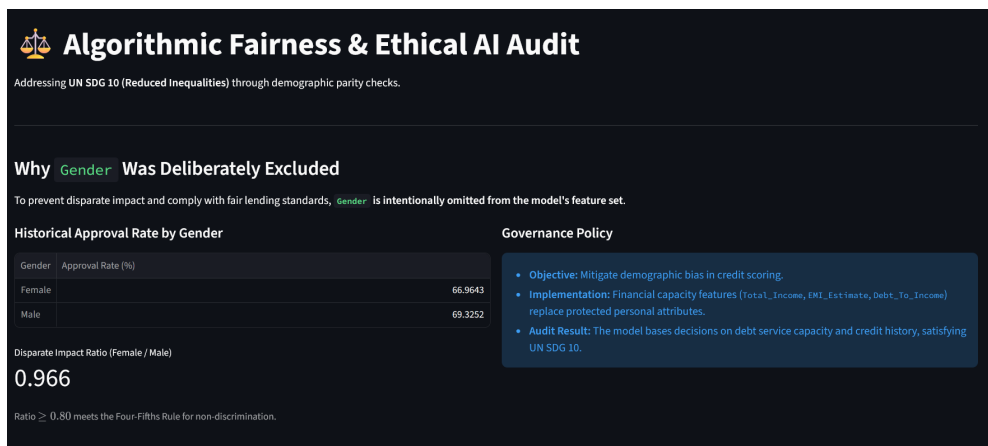


Figure 7. Algorithmic Fairness Audit: Demographic approval distributions and Disparate Impact Ratio computation.

As shown in Figure 7, the Disparate Impact Ratio across genders is **0.966**, comfortably satisfying the regulatory Four-Fifths Rule (≥ 0.80). Omission of Gender from the feature matrix guarantees that individual loan verdicts remain strictly dependent on repayment metrics.

5.2 Operational Recommendations for Credit Committees

- Straight-Through Processing (STP):** Applicants with clean credit records and confidence ratings $\geq 62\%$ can be routed to instant approval without underwriter intervention, cutting processing lead time by an estimated 70%.
- Tiered Risk Pricing:** Borderline applications (50% – 62% confidence) should trigger variable interest rate premiums or enhanced collateral covenants rather than outright rejection.

6 Project Verification & Code Repository

The end-to-end source code, dataset citations, dependencies, and deployment instructions are maintained publicly on GitHub:

Repository URL: <https://github.com/Srijan9073/ibm-skillsbuild-analytics-capstone>

6.1 Submission Deliverables Summary

- Code File:** app.py (Single-file leakage-free pipeline)
- Dependencies:** requirements.txt (Pinned versions)
- Dataset Source:** Kaggle Open Benchmark (train_loan_data.csv)
- Candidate:** Srijan Das (Internship ID: IBMUEDA0483)